

Pythagoras, mensuration and 3D - Set 2 - Solutions

Specification M5.7, M5.11-M5.13, M5.15-M5.17.

Answer key

Question	Answer	Topic
1	C	Pythagoras, mensuration and 3D
2	D	Pythagoras, mensuration and 3D
3	D	Pythagoras, mensuration and 3D
4	D	Pythagoras, mensuration and 3D
5	C	Pythagoras, mensuration and 3D

Worked solutions

Question 1 MEN-16 · Pythagoras, mensuration and 3D · spec M5.18

From the top of a cliff 40 m high, the angle of depression of a boat is 30° . How far is the boat from the foot of the cliff?

Answer: C

$$\tan 30^\circ = \frac{40}{d} \Rightarrow d = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}.$$

Full method - how and why

An angle of depression from the top equals the angle of elevation from the boat. Then it is right-angled trigonometry.

- 1 Draw the triangle: the cliff is vertical (40 m), the sea horizontal, and the angle at the boat is 30° .
why: The angle of depression from the cliff top equals the angle of elevation from the boat (alternate angles with the horizontal).
- 2 Relative to the 30° angle, the cliff is opposite and the distance d is adjacent: $\tan 30^\circ = \frac{40}{d}$.
why: $\tan = \frac{\text{opposite}}{\text{adjacent}}$.
- 3 Solve: $d = \frac{40}{\tan 30^\circ} = \frac{40}{1/\sqrt{3}} = 40\sqrt{3}$ m.
why: Dividing by $\frac{1}{\sqrt{3}}$ multiplies by $\sqrt{3}$.

Where it goes wrong

- Multiplying by $\tan 30^\circ$ instead of dividing, giving $\frac{40}{\sqrt{3}}$ - an offered option.
- Placing the 30° angle at the top of the cliff inside the triangle, which would swap opposite and adjacent.

Takeaway Depression from the top = elevation from the bottom; a shallow angle means a large horizontal distance.

Question 2 X2-29 · Pythagoras, mensuration and 3D · spec M5.16

P : 'a solid cylinder and a solid cube have equal volumes'. Q : 'the cylinder and the cube have equal surface areas'. Which one of the following is true?

Answer: D

Equal volumes say nothing about surface areas (a long thin cylinder has a large surface area), and equal surface areas say nothing about volumes. Neither implies the other.

Full method - how and why

Volume and surface area are independent quantities. Neither condition constrains the other, so test both directions.

- 1 Not sufficient: take a cube of side 1 (volume 1, surface 6) and a cylinder of volume 1 that is long and thin - say radius 0.1, height $\frac{1}{0.01\pi} \approx 31.8$, whose surface area is far larger than 6.
why: Equal volumes with wildly different shapes give wildly different surface areas.
- 2 Not necessary: two solids can share a surface area while having different volumes, by the same flexibility of shape.
why: Stretching a cylinder thin at fixed surface area shrinks its volume.
- 3 So P is neither necessary nor sufficient for Q .
why: Match to the list.

Where it goes wrong

- Assuming 'bigger volume means bigger surface area' - among different shapes, no such rule holds.
- Testing only one direction and then guessing the other.

Takeaway Volume and surface area are independent across shapes; a sphere minimises surface area for a given volume, and everything else can be made worse without limit.

Question 3 MEN-02 · Pythagoras, mensuration and 3D · spec M5.16

A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.

Answer: D

$$\pi r^2 \cdot 2r = 2\pi r^3 = 54\pi \Rightarrow r = 3, h = 6. \text{ Surface area} = 2\pi r^2 + 2\pi rh = 18\pi + 36\pi = 54\pi.$$

Full method - how and why

Use the volume condition to find r , then substitute into the surface-area formula. Both formulas must be recalled.

- 1 Write the volume with $h = 2r$: $V = \pi r^2 h = \pi r^2 (2r) = 2\pi r^3$.
why: Substituting the relationship immediately reduces to one unknown.
- 2 Solve $2\pi r^3 = 54\pi$: $r^3 = 27$, so $r = 3$ and $h = 6$.
why: The π cancels, which is a good sign.
- 3 Total surface area $= 2\pi r^2$ (two circular ends) $+ 2\pi rh$ (curved surface).
why: 'Total' means include both ends; 'curved surface area' would be just $2\pi rh$.
- 4 Substitute: $2\pi(9) + 2\pi(3)(6) = 18\pi + 36\pi = 54\pi$.
why: Coincidentally equal to the volume's numerical value - a deliberate touch.

Where it goes wrong

- Giving only the curved surface 36π - an offered option.
- Forgetting that there are **two** circular ends.

Takeaway Cylinder: $V = \pi r^2 h$, curved surface $2\pi rh$, total surface $2\pi r^2 + 2\pi rh$.

Question 4 MEN-10 · Pythagoras, mensuration and 3D · spec M5.7

A square-based pyramid has base side 6 and all slant edges of length 9. Find its height.

Answer: D

$$\text{Half the base diagonal is } 3\sqrt{2}, \text{ so } h^2 = 81 - 18 = 63.$$

Full method - how and why

The height of a pyramid, a slant edge and half the base diagonal form a right-angled triangle.

- 1 Find the base diagonal: a square of side 6 has diagonal $6\sqrt{2}$.
why: Pythagoras on the square base: $\sqrt{36 + 36} = 6\sqrt{2}$.
- 2 Half of it is $3\sqrt{2}$ - the horizontal distance from the centre to a corner.
why: The apex sits above the centre of the base, so this is the horizontal leg.
- 3 Apply Pythagoras with the slant edge as hypotenuse: $h^2 = 9^2 - (3\sqrt{2})^2 = 81 - 18 = 63$.
why: $(3\sqrt{2})^2 = 9 \times 2 = 18$ - square the coefficient too.
- 4 So $h = \sqrt{63}$.
why: $= 3\sqrt{7}$, if you prefer it simplified.

Where it goes wrong

- Using half the **side** (3) instead of half the diagonal, giving $\sqrt{72}$ - an offered option.
- Computing $(3\sqrt{2})^2$ as 6 or 9.

Takeaway For a slant **edge**, use half the base diagonal; for a slant **face height**, use half the base side.

Question 5 MEN-13 · Pythagoras, mensuration and 3D · spec M5.7

The points $A(1, 2)$, $B(4, 6)$ and $C(k, 2)$ with $k > 1$ form a triangle with a right angle at B . Find k .

Answer: C

Gradient $AB = \frac{4}{3}$, so BC must have gradient $-\frac{3}{4}$: $\frac{2-6}{k-4} = -\frac{3}{4} \Rightarrow 16 = 3(k-4) \Rightarrow k = \frac{28}{3}$.

Full method - how and why

A right angle in coordinate geometry means perpendicular gradients - the vertex of the right angle is the shared point.

- 1 Compute the gradient of AB : $\frac{6-2}{4-1} = \frac{4}{3}$.
why: The right angle is at B , so the two sides meeting there are BA and BC .
- 2 Perpendicular gradient for BC : $-\frac{3}{4}$.
why: Negative reciprocal.
- 3 Write the gradient of BC using $C(k, 2)$: $\frac{2-6}{k-4} = \frac{-4}{k-4}$, and set it equal to $-\frac{3}{4}$.
why: Keep the subtraction order consistent: from B to C in both coordinates.
- 4 Solve: $\frac{-4}{k-4} = -\frac{3}{4} \Rightarrow 16 = 3(k-4) \Rightarrow k = 4 + \frac{16}{3} = \frac{28}{3}$.
why: Cross-multiply; both sides are negative so the signs cancel.

Where it goes wrong

- Assuming the right angle is at A or C .
- Dropping the $+4$ at the end and answering $\frac{16}{3}$ - an offered option.

Takeaway Right angle at a named vertex: use the two sides **through that vertex** and set the gradient product to -1 .